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Unraveling effects of foliar protein hydrolysates on growth, physiology and water-use efficiency of greenhouse tomato and common bean

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Abstract

The growing demand for sustainable crop management has intensified interest in protein hydrolysates (PHs) as bioactive biostimulants. While the efficacy of foliar PH applications has been documented in tomato at concentrations of 2.5 mL/L and above, or through combined foliar-drench programs, evidence for significant agronomic responses at low foliar rates- particularly in legumes- remains limited. The present study evaluated the effects of two novel plant- and microbial-derived PH formulations, AM2025 and MP2025, applied twice as foliar sprays at 0.05% and 0.1% (v/v) (equivalent to 0.5 and 1.0 mL/L) on vegetative growth, photosynthesis and water-use efficiency in *Phaseolus vulgaris* L. and *Solanum lycopersicum* L. under controlled greenhouse conditions. In *S. lycopersicum*, several morphological, biomass, and physiological traits, including leaf area, shoot biomass, leaf chlorophyll content (SPAD), net photosynthetic rate, and water-use efficiency, were significantly affected by PH treatments. A strong interaction between formulation and application rate was observed, indicating that treatment effects depended on their specific combination. Notably, these responses were achieved with only two foliar applications at concentrations lower than those typically reported in the literature. In *P. vulgaris*, responses were more limited and variable, with fewer statistically significant effects. However, some parameters, including shoot dry weight, leaf area, and intrinsic water-use efficiency, showed treatment-related increases. These findings demonstrate species-specific responses to PH application and highlight the importance of formulation-dose interactions, while also suggesting that these products exhibit considerable biological activity at very low application rates. Further research is needed to validate these effects across a wider range of species, application regimes, and field conditions.

Keywords Biostimulants, Amino acids, Protein hydrolysates, Tomato, Common bean, Foliar application, Photosynthesis, Water-use efficiency

1 Introduction

The increasing demand for agricultural productivity under climate change and resource limitations has intensified the need for sustainable crop management strategies [1, 2]. Intensive reliance on synthetic fertilizers has significantly contributed to greenhouse



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gas emissions, nutrient leaching, and broader soil and ecosystem degradation, thereby threatening agroecosystem sustainability [3, 4]. In this context, improving nutrient use efficiency through optimized fertilizer management, reduced synthetic input rates, and the use of biostimulants represents key strategies for maintaining productivity while lowering environmental burdens [3–5]. Among these alternatives, PHs and other biostimulants derived from agro-industrial residues have emerged as promising tools, promoting resource efficiency, waste valorization, and a reducing dependence on non-renewable inputs [5–7]. In the EU, biostimulants were formally defined and harmonized as a distinct category of fertilising products [8]. Despite increasing research interest in biostimulants, important gaps remain in understanding their mechanisms of action and standardizing application strategies across crops and environments [5, 6, 9].

Under EU Regulation 2019/1009, plant biostimulants are defined as products that stimulate plant nutrition processes independently of their nutrient content, aiming to improve nutrient use efficiency, abiotic stress tolerance, crop quality, or nutrient availability in the rhizosphere [8]. Among them, protein hydrolysates (PHs) represent one of the most extensively studied categories [7]. These products, composed of free amino acids and short peptides, can act as signaling and metabolic modulators, influencing nitrogen assimilation and enzyme activities involved in plant metabolism [10, 11]. When applied as foliar treatments, PHs have been shown to enhance key physiological processes, including chlorophyll content, stomatal function, and photosynthetic performance [12, 13]. Similar physiological effects have also been reported for amino acid-based biostimulants. Exogenous amino acid applications in vegetable crops have been shown to enhance plant growth, nutrient accumulation, and photosynthetic assimilation, suggesting that amino acids and small peptides can contribute to improved metabolic activity and resource-use efficiency in plants [14, 15].

Improvements in net photosynthetic rate and intrinsic water use efficiency have been associated with physiological adjustments involving nitrogen metabolism and stomatal regulation under both optimal and stress conditions [7, 16]. Moreover, foliar PH application may enhance biomass accumulation and vegetative growth, potentially through hormone-like effects and improved nutrient uptake efficiency [5, 17].

Previous studies have demonstrated beneficial effects of protein hydrolysates (PHs) on horticultural and legume crops, including tomato and common bean, reporting improvements in vegetative growth, root development, nutrient uptake, and physiological performance related to photosynthesis [5, 7, 12]. In tomato, PH application has been associated with increased biomass production, enhanced leaf greenness (SPAD), and improved nitrogen metabolism, whereas in legumes, improvements in root growth and nitrogen assimilation have been reported [7]. However, plant responses vary considerably depending on PH origin and composition, application rate, and crop species. Plant- and animal-derived PHs differ in peptide composition, amino acid profiles, and associated bioactive compounds, potentially eliciting distinct physiological and metabolic responses, although mechanistic evidence remains limited [5, 6]. Moreover, dose-dependent responses are common, with optimal concentrations promoting growth and, in some cases, stress tolerance, while excessive rates may cause neutral or inhibitory effects [5, 7]. Crop-specific traits, including nitrogen metabolism, growth dynamics, and developmental sensitivity, further contribute to variability in responsiveness. In addition, emerging evidence suggests that plant responses to PHs are not always proportional to

the applied dose, and relatively low concentrations may be sufficient to trigger physiological responses, potentially related to the activity of bioactive peptides [10, 18, 19]. Nevertheless, foliar applications of protein hydrolysates in experimental studies are generally performed at low to moderate concentrations (approximately 2–5 mL L⁻¹ in experimental studies), particularly in tomato and related horticultural crops [7, 12]. Consequently, the potential for achieving satisfactory agronomic responses at reduced application rates remain insufficiently explored, particularly in leguminous species, where responses to biostimulants are less extensively characterized and may be more variable [5, 19].

Despite growing evidence supporting the efficacy of PHs, comparative studies assessing different PH types and reduced foliar application rates under standardized conditions remain limited. Cherry tomato and common bean were selected as model crops due to their rapid growth and suitability for evaluating early vegetative responses under controlled greenhouse conditions. This approach enables efficient comparison of PH formulations prior to field-scale validation [5].

It was hypothesized that PH formulations enriched in small peptides (< 3 kDa) would enhance vegetative growth and physiological performance even at limited and low foliar application timing and rates. The objective of this study was to evaluate the effects of such an application strategy on the vegetative growth, photosynthetic performance, biomass accumulation, and water-use efficiency of *Phaseolus vulgaris* L. and *Solanum lycopersicum* L. under controlled greenhouse conditions. For this purpose, two small-peptide-enriched products were used: one plant-derived (MP2025) and one microbial-derived (AM2025). Specifically, this study addressed the following questions: (1) Do both small-peptide protein hydrolysate formulations improve vegetative growth and water-use efficiency at very low foliar application timing and rates? and (2) How do their effects vary across increasing, yet reduced, foliar concentrations?

2 Materials and methods

2.1 Biostimulant products

Two PH formulations were provided by Croptivate BV (Maastricht, The Netherlands) and designated as AM2025 and MP2025. According to the manufacturer, MP2025 is a plant-derived (*Zea mays*) enzymatically hydrolyzed formulation, while AM2025 is a microbial-derived (*Saccharomyces cerevisiae*) formulation. Both products consist of peptide- and amino acid-containing hydrolysates and were tested as foliar-applied biostimulants.

2.2 Plant material and growth conditions

Two separate greenhouse experiments were carried out in 2025 at the University of Bonn using common bean (*Phaseolus vulgaris* L., cv. Saxa) and cherry tomato (*Solanum lycopersicum* L., cv. Sungold). These cultivars were selected because they are widely cultivated and commonly used in horticultural production and research systems and exhibit rapid early growth.

Seeds were sown in plastic pots filled with a cocopeat-perlite substrate (70:30 v/v). Pots of 8 cm diameter were used for common bean, and 12 cm diameter pots for cherry tomato. Three seeds were sown per pot at a depth of approximately 0.5 cm, and seedlings were thinned to one uniform plant per pot after emergence to ensure consistent plant density. Plants were grown under controlled greenhouse conditions with day/night

temperatures of 22/18°C, relative humidity of 60%, and a photoperiod of 12 h (common bean) and 14 h (cherry tomato). Light intensity (photosynthetically active radiation, PAR) was maintained at approximately $400 \mu\text{mol m}^{-2} \text{s}^{-1}$ using the greenhouse control system, ensuring consistent illumination throughout the experimental period. Irrigation was carried out using Hoagland nutrient solution. Plants were supplied with half-strength solution during early vegetative growth (up to 10 days after sowing, DAS) and full-strength solution thereafter. The common bean experiment was carried out from October 10 to November 4, 2025 (28 days), while the cherry tomato experiment was conducted from November 6 to December 25, 2025 (49 days).

2.3 Experimental design and treatments

Both experiments were arranged in a factorial completely randomized design (CRD). Each treatment consisted of eight biological replicates (individual plants). The experimental factors were PH formulation (AM2025 and MP2025) and foliar application concentrations (0%, 0.05%, and 0.1% v/v). The untreated control (0%) was sprayed with water. Foliar applications were applied twice during the vegetative stage: at the appearance of the first true leaves (13 DAS for common bean and 20 DAS for cherry tomato) and again seven days later. Solutions were applied using a handheld sprayer at a volume equivalent to 600 L ha^{-1} , corresponding to approximately 1 mL per plant, to ensure uniform leaf coverage.

2.4 Plant growth measurements

At the end of the experimental period, all plants were harvested to assess vegetative growth. Leaf greenness (SPAD index) used as a proxy for chlorophyll content, shoot fresh weight (SFW, g plant^{-1}), root fresh weight (RFW, g plant^{-1}), shoot dry weight (SDW, g plant^{-1}), root dry weight (RDW, g plant^{-1}), and leaf area (LA, $\text{cm}^2 \text{ plant}^{-1}$) were measured for both species. Roots were carefully washed under running water to remove soil and other debris, blotted gently with tissue paper to remove surface moisture, and weighed to determine fresh weight. Dry weights were obtained after oven-drying the samples at 105°C for 24 h until constant weight was achieved. Leaf area was measured using a leaf area meter (LI-3100 C, LI-COR Biosciences, Lincoln, Nebraska, USA). In cherry tomato, additional morphological traits were recorded: plant height (cm) from the first stem node to the tip of the last fully expanded leaf, fourth leaf extension (LE4, cm) from the main branch to the tip of the fourth leaf, and stem diameter (mm) at the first internode above the base using a digital caliper.

Leaf greenness, used as a proxy for chlorophyll status, was measured on the third fully expanded leaf from the apex during the vegetative growth stage using a SPAD-502 m (Minolta, Japan). Three measurements were taken per leaf and averaged to obtain a single SPAD value, with eight biological replicates per treatment. Leaf gas exchange was measured on the third fully expanded leaf from the plant apex using a portable infrared photosynthesis system (6400XT, LI-COR Biosciences, Lincoln, Nebraska, USA) under controlled conditions with a reference CO_2 concentration of 400 ppm, photosynthetic photon flux density of $1000 \mu\text{mol m}^{-2} \text{s}^{-1}$, and a flow rate of $400 \mu\text{mol s}^{-1}$ at the ambient greenhouse leaf temperature (22°C). Measurements were taken on eight replicate plants per treatment. The following variables were recorded: leaf net photosynthetic rate (Pn,

$\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$), stomatal conductance (g_s , $\text{mol H}_2\text{O m}^{-2} \text{ s}^{-1}$), transpiration rate (E , $\text{mmol H}_2\text{O m}^{-2} \text{ s}^{-1}$), and intercellular CO_2 concentration (C_i , ppm).

Intrinsic water use efficiency (WUEi) and instantaneous water use efficiency (WUE) were calculated according to the following equations [20]:

$$WUE = \frac{Pn}{E} \quad \text{and} \quad WUE? = \frac{Pn}{g_s}$$

With:

Pn = leaf net photosynthetic rate ($\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$).

E = transpiration rate ($\text{mmol H}_2\text{O m}^{-2} \text{ s}^{-1}$).

g_s = stomatal conductance ($\text{mol H}_2\text{O m}^{-2} \text{ s}^{-1}$).

2.5 Statistical analysis

Data from the common bean and cherry tomato experiments were analyzed separately. For each species, measured variables were subjected to two-way analysis of variance (ANOVA) using a factorial model to evaluate the effects of PH formulation, application rates and their interaction. The experimental unit was a single plant ($n=8$ per treatment).

When significant main or interaction effects were detected ($p \leq 0.05$), mean separation was performed using Duncan's multiple range test (DMRT) at the 5% probability level. All statistical analyses were performed using R software (R version 4.5.2; R Core Team, 2025).

3 Results

3.1 Overall effects of PH formulations and application rate across both crops

The analysis of variance showed that the effects of PH formulations, application rate, and their interaction varied by crop species and measured parameter (Tables 1 and 2). Overall, application rate was the main driver of plant responses, while PH formulation alone showed limited significant effects in both species.

In *P. vulgaris*, no significant effects of PH formulation were detected. Application rate significantly affected shoot dry weight, net photosynthetic rate, stomatal conductance, and intrinsic water-use efficiency, while PH $\times$ Dose interactions were limited to leaf area and transpiration. In *S. lycopersicum*, more pronounced responses were observed. Application rate significantly influenced most physiological and growth traits, including SPAD, leaf area, photosynthesis, transpiration, intercellular CO_2 concentration, and water-use efficiency. Significant PH $\times$ Dose interactions were also found for transpiration and WUE, indicating formulation-dependent responses at specific application rates.

Overall, cherry tomato showed stronger and more consistent sensitivity to PH treatments than common bean, with application rate representing the dominant factor regulating plant responses.

3.1.1 Effect of PH formulations on vegetative growth and water-use efficiency

To evaluate whether the two PH formulations improved vegetative growth and water-use efficiency, their main effects were assessed across species. Non-significant differences are described without statistical inference.

Table 1 Analysis of variance (p-values) showing the effects of PH formulation, application rate (Dose), and their interaction on physiological and growth parameters in *P. vulgaris* (n = 8)

Source of variation	DF	SPAD	SFW	RFW	SDW	RDW	LA	Pn	g _s	E	CI	WUE	WUEi
PH	1	0.38	0.90	0.87	0.25	0.97	0.34	0.38	0.68	0.96	0.31	0.69	0.57
Dose	2	0.60	0.20	0.31	0.01*	0.16	0.10	0.74	0.04*	0.38	0.05*	0.61	0.05*
PH x Dose	2	0.28	0.36	0.25	0.93	0.85	0.04*	0.08	0.07	0.05	0.21	0.57	0.19

Table 2 Analysis of variance (p-values) showing the effects of PH formulation, application rate (Dose), and their interaction on physiological (A) and growth (B) parameters in *S. lycopersicum* ($n=8$)**(A) Physiological parameters**

Source of variation	DF	SPAD	LA	Pn	g_s	E	Ci	WUE	WUE _i
PH	1	0.512	0.721	0.718	0.261	0.111	0.280	0.029*	0.190
Dose	2	0.016*	0.030*	0.035*	0.580	0.043*	0.009**	0.007**	0.180
PH × Dose	2	0.428	0.336	0.260	0.133	0.030*	0.354	< 0.001***	0.398

(B) Growth parameters

Source of variation	DF	PH	LE	SD	SFW	RFW	SDW	RDW
PH	1	0.943	0.310	0.974	0.630	0.787	0.641	0.798
Dose	2	< 0.001***	< 0.001***	0.012*	0.195	0.005**	0.202	0.003**
PH × Dose	2	0.007**	0.480	0.503	0.959	0.980	0.964	0.971

Values represent p-values. Significance levels: *** $p \leq 0.001$, ** $0.001 < p \leq 0.01$, * $0.01 < p \leq 0.05$

SPAD=leaf greenness index; SFW=shoot fresh weight; RFW=root fresh weight; SDW=shoot dry weight; RDW=root dry weight; LA=leaf area; Pn=net photosynthetic rate; g_s =stomatal conductance; E=transpiration rate; Ci=intercellular CO₂ concentration; WUE=water-use efficiency; WUE_i=intrinsic water-use efficiency; PH= plant height, LE= leaf expansion and SD= stem diameter

3.2 Water-use efficiency

In *S. lycopersicum*, water-use efficiency (WUE) was significantly affected by PH application (Fig. 1A). The highest WUE was observed with MP2025 at 0.1%, representing an increase of approximately 31.8% compared to the control. AM2025 at 0.05% also increased WUE, whereas no improvement was observed at 0.1%. Intrinsic water-use efficiency (WUE_i) in *S. lycopersicum* showed limited variation among treatments; however, MP2025 at 0.05% resulted in the highest value and was significantly greater than AM2025 at 0.1% (Fig. 1C). In *P. vulgaris*, WUE was not significantly affected by PH application (Fig. 1B). Intrinsic water-use efficiency (WUE_i), however, differed among treatments, with higher values observed for AM2025 at 0.1% and MP2025 at 0.05% compared to the control (Fig. 1D).

3.2.1 Effect of PH application rates on vegetative growth parameters and water-use efficiency

The effects of application rate and the interaction between PH formulation and application rate were examined for parameters showing significant responses.

3.3 Leaf greenness (SPAD)

In cherry tomato, both PH formulations increased leaf greenness compared to the control, with a significant effect observed for MP2025 at 0.1% (Fig. 2A). In common bean, leaf greenness was not significantly affected by the treatments (Fig. 2B), although MP2025 at 0.1% showed a positive numerical trend (~7.3% above the control).

3.3.1 Photosynthetic performance

In cherry tomato, net photosynthetic rate (Pn) was enhanced at the 0.05% application rate, with a statistically significant increase observed only for AM2025 (approximately +16.5% compared to the control), while MP2025 showed no significant difference (Fig. 3A). In common bean, Pn remained statistically unchanged across treatments, although MP2025 at 0.1% showed a slight, non-significant increase compared to the control (Fig. 3B).

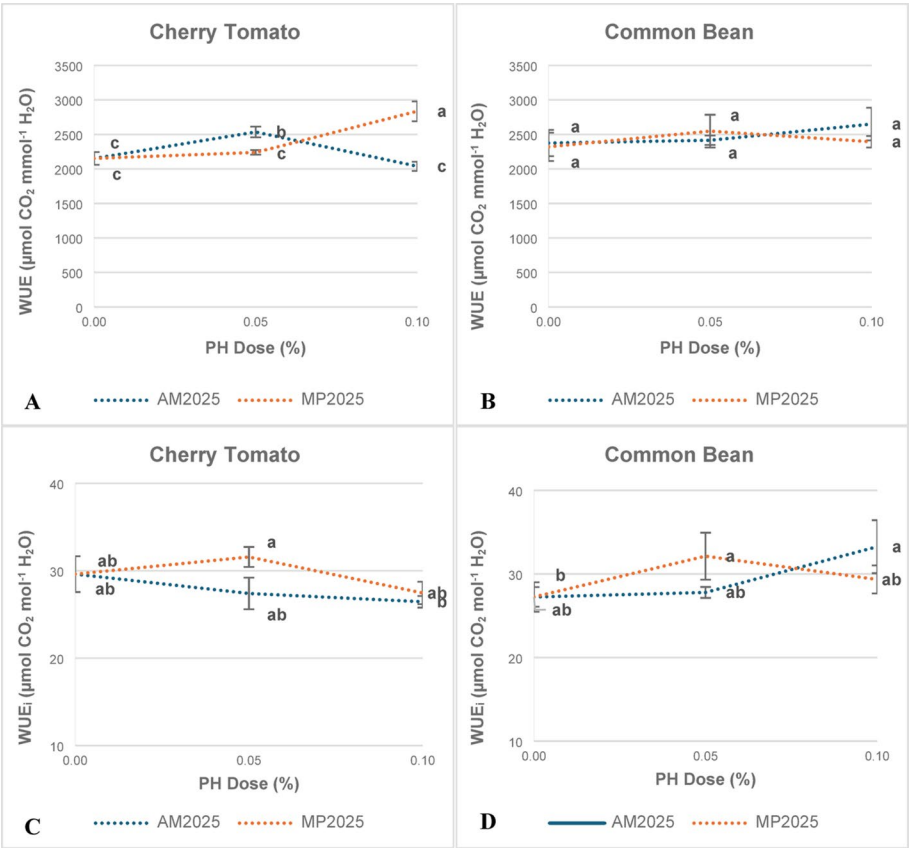


Fig. 1 Effect of PH Formulations (MP2025 and AM2025) and application rates (0, 0.05%, and 0.1%) on water-use efficiency in *S. lycopersicum* and *P. vulgaris*. **A** Water-use efficiency (WUE, $\mu\text{mol CO}_2 \text{ mmol}^{-1} \text{ H}_2\text{O}$) in *S. lycopersicum*; **B** WUE in *P. vulgaris*; **C** intrinsic water-use efficiency (WUE_i, $\mu\text{mol CO}_2 \text{ mol}^{-1} \text{ H}_2\text{O}$) in *S. lycopersicum*; **D** WUE_i in *P. vulgaris*. Values represent mean $\pm$ standard error ($n=8$). Different small letters indicate significant differences according to Duncan's multiple range test ($p \leq 0.05$)

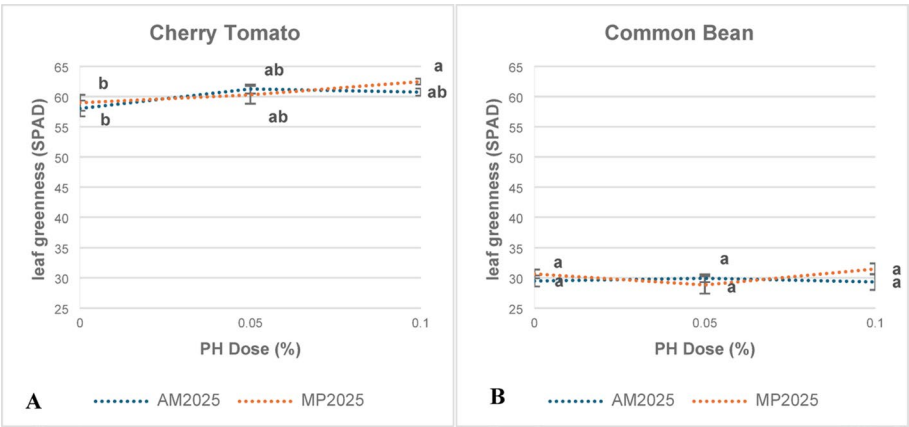


Fig. 2 Effect of PH Formulations (MP2025 and AM2025) and application rates (0, 0.05% and 0.1%) on leaf greenness (SPAD) in **(A)** *S. lycopersicum* and **(B)** *P. vulgaris*. Values represent mean $\pm$ standard error ($n=8$). Different small letters indicate significant differences according to Duncan's multiple range test ($p \leq 0.05$)

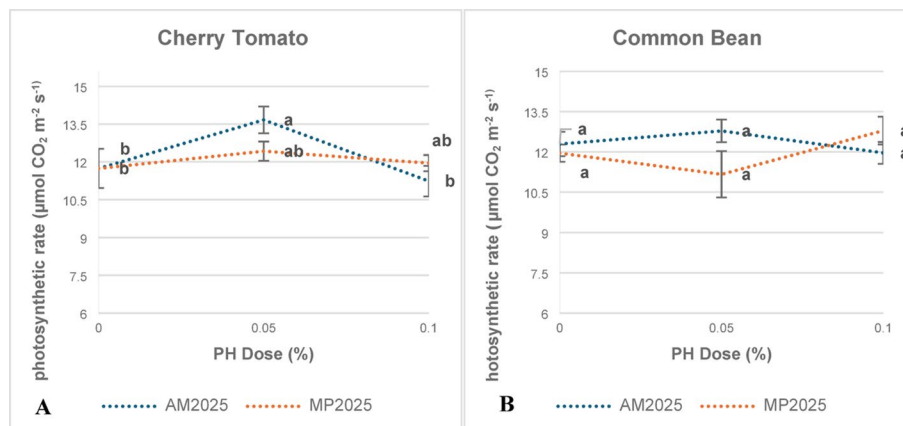


Fig. 3 Effect of PH Formulations (MP2025 and AM2025) and application rates (0, 0.05%, and 0.1%) on net photosynthetic rate (P_n , $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$) in (A) *S. lycopersicum* and (B) *P. vulgaris*. Values represent mean $\pm$ standard error ($n=8$). Different small letters indicate significant differences according to Duncan's multiple range test ($p \leq 0.05$)

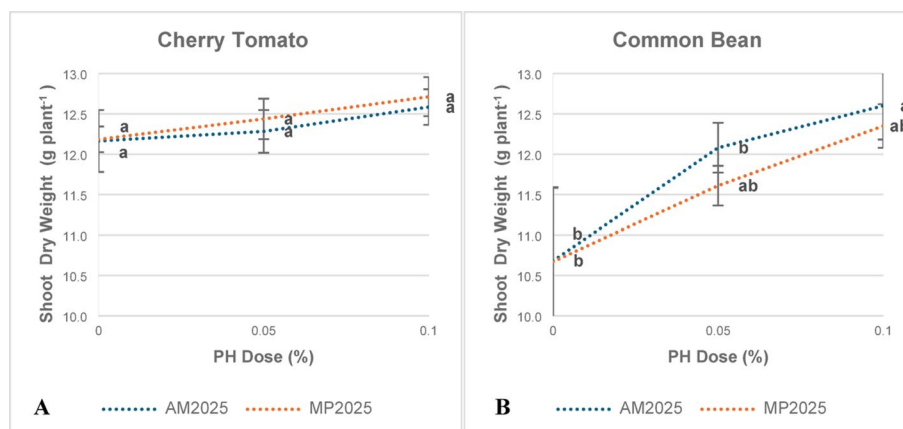


Fig. 4 Effect of PH Formulations (MP2025 and AM2025) and application rates (0, 0.05%, and 0.1%) on shoot dry weight (SHDW, g plant^{-1}) in (A) *S. lycopersicum* and (B) *P. vulgaris*. Values represent mean $\pm$ standard error ($n=8$). Different small letters indicate significant differences according to Duncan's multiple range test ($p \leq 0.05$)

3.4 Biomass accumulation

Shoot dry weight was significantly influenced by application rate, whereas no significant main effect of PH formulation was detected (Fig. 4). In cherry tomato, both formulations resulted in numerical increases in shoot biomass, with a slightly greater trend observed for MP2025 (Fig. 4A). In common bean, only AM2025 at 0.1% significantly increased shoot dry weight by approximately 18% compared with the control (Fig. 4B).

3.4.1 Leaf Area

Leaf area was significantly affected by PH application and dose in both species (Fig. 5). In cherry tomato, AM2025 at 0.05% resulted in the highest leaf area values (Fig. 5A). In common bean, the largest increase was observed with AM2025 at 0.05%, resulting in approximately +32.9% compared to the control (Fig. 5B).

3.5 Transpiration

Leaf transpiration (E) showed treatment-dependent responses (Fig. 6). In cherry tomato, MP2025 at 0.1% significantly reduced transpiration compared to the control (Fig. 6A). In

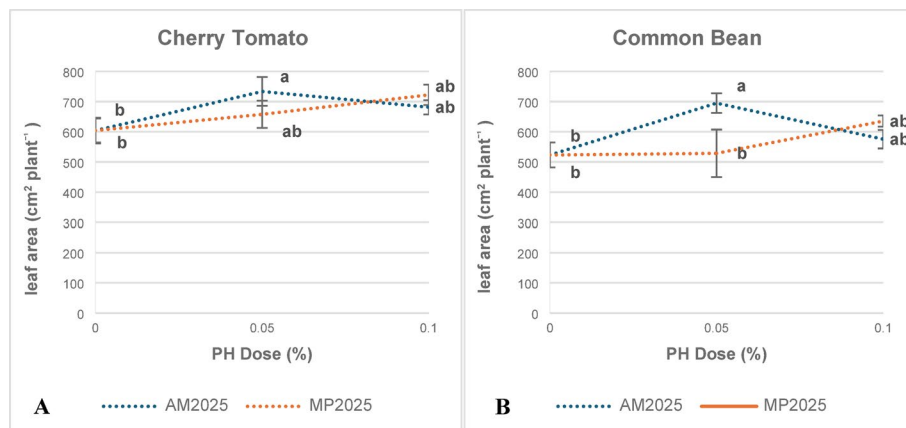


Fig. 5 Effect of PH Formulations (MP2025 and AM2025) and application rates (0, 0.05%, and 0.1%) on leaf area (LA, $\text{cm}^2 \text{ plant}^{-1}$) in (A) *S. lycopersicum* and (B) *P. vulgaris*. Values represent mean $\pm$ standard error ($n=8$). Different small letters indicate significant differences according to Duncan's multiple range test ($p \leq 0.05$)

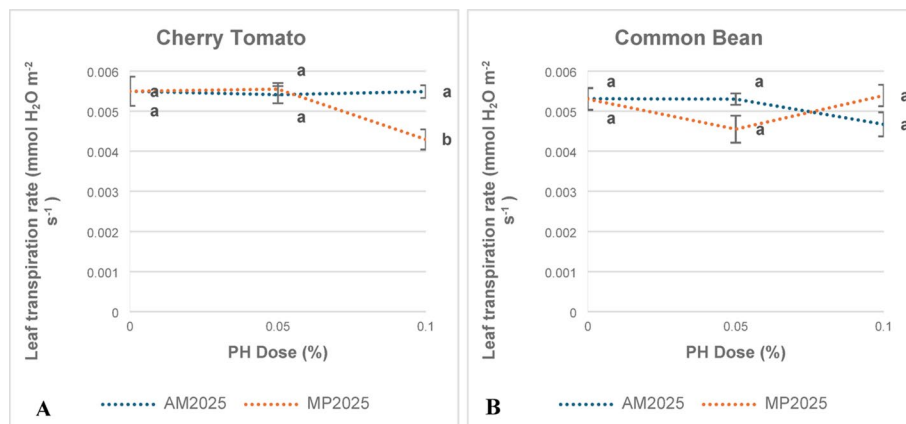


Fig. 6 Effect of PH Formulations (MP2025 and AM2025) and application rates (0, 0.05%, and 0.1%) on leaf transpiration rate (E, $\text{mmol H}_2\text{O m}^{-2} \text{ s}^{-1}$) in (A) *S. lycopersicum* and (B) *P. vulgaris*. Values represent mean $\pm$ standard error ($n=8$). Different small letters indicate significant differences according to Duncan's multiple range test ($p \leq 0.05$)

common bean, transpiration showed no significant differences, although slight numerical variations were observed (Fig. 6B).

Overall, both PH formulations promoted biomass accumulation and canopy development, with stronger responses observed in cherry tomato than in common bean.

3.6 Additional morphological traits in cherry tomato

In cherry tomato, PH treatments significantly improved early vegetative traits, including plant height, leaf expansion, and stem diameter (Supplementary Fig. S1). The low application rates (0.05–0.1%) were sufficient to achieve these effects, highlighting the efficiency of low-dose applications.

4 Discussion

4.1 Effect of PH formulations on growth and water-use efficiency

This study investigated whether foliar application of small-peptide protein hydrolysates (PHs) improves vegetative growth and water-use efficiency (WUE) in cherry tomato (*S. lycopersicum*) and common bean (*P. vulgaris*) at very low application rates. Overall,

PH application enhanced plant growth and physiological performance, with responses strongly dependent on crop species.

In *S. lycopersicum*, PH treatments improved leaf greenness, indicating enhanced chlorophyll content and suggesting improved photosynthetic performance. This is consistent with previous reports showing increased chlorophyll-related traits and modulation of nitrogen metabolism [12, 21]. Similar improvements in photosynthetic activity, nutrient acquisition, and biomass production have been reported in vegetable crops following amino acid-based biostimulant applications. For example, amino acid supplementation in lettuce enhanced photosynthetic assimilation and nutrient availability, while recent studies also reported improvements in growth and nutrient content following amino acid application [14, 15]. These findings support the hypothesis that amino acids and small peptides present in protein hydrolysates may contribute to the observed improvements in plant physiological performance. The increase in net photosynthetic rate observed at 0.05%, particularly under AM2025, further indicates improved carbon assimilation efficiency. Biomass accumulation and leaf area were also positively affected in tomato, suggesting improved growth potential. Increased leaf area likely enhanced light interception, contributing to improved growth and biomass production [7]. Water-use efficiency was significantly improved in tomato, particularly with MP2025 at 0.1%. This response was associated with enhanced photosynthetic performance and reduced transpiration, without changes in stomatal conductance, indicating improved carbon gain per unit water loss rather than a primary effect mediated by stomatal limitation. This mechanism is consistent with studies showing that protein hydrolysates enhance photosynthetic performance and improve water-use efficiency, often associated with coordinated changes in carbon assimilation and plant water relations [12, 19, 22, 23]. In contrast, *P. vulgaris* showed limited responses in growth and photosynthesis. However, PH application reduced stomatal conductance, leading to increased intrinsic water-use efficiency (WUE_i). This suggests that water-saving in bean mainly associated with stomatal regulation rather than enhanced carbon assimilation, consistent with general drought-adaptive stomatal regulations strategies in legumes and other crops [24, 25].

The different responses observed between the two PH formulations may be related, at least in part, to their distinct biological origins. MP2025 is derived from maize proteins, whereas AM2025 originates from *Saccharomyces cerevisiae*. Previous studies have shown that the biostimulant activity of protein hydrolysates depends on the source material and hydrolysis process, which influence their biochemical composition and biological activity in plants [7, 19]. Therefore, the contrasting responses observed between AM2025 and MP2025 may reflect differences in the bioactive compounds generated during hydrolysis, resulting in formulation-specific effects on plant growth and water-use efficiency. Similar formulation-dependent effects have been reported for vegetal protein hydrolysates, where application strategy and nitrogen conditions influenced lettuce growth and composition, further demonstrating that PH effectiveness depends on product characteristics and growing conditions [26].

Overall, these results support the first hypothesis that PH formulations can enhance vegetative growth and water-use efficiency at very low foliar application rates, with stronger responses observed in tomato than in bean.

4.2 Effect of foliar application rate on growth and water-use efficiency

This study also examined how plant responses vary across increasing but low foliar PH concentrations (0, 0.05%, and 0.1%). Application rate played a key role in determining plant responses, with moderate concentrations generally producing the most consistent physiological effects.

In *S. lycopersicum*, both 0.05% and 0.1% treatments improved physiological performance, with 0.05% showing stronger effects on photosynthesis and 0.1% showing stronger effects on water-use efficiency. This suggests a dose-dependent shift in PH action from stimulation of carbon assimilation at lower concentration to improved water-use efficiency at higher concentration. In *P. vulgaris*, responses were weaker and mainly restricted to stomatal regulation at higher concentration, indicating lower sensitivity of this species to foliar PH application. These findings support the concept that PHs are often considered bioactive compounds influencing metabolic and physiological pathways [5, 17, 19]. The observed effects at relatively low application rates further suggest a high biological activity of the tested formulations. Overall, the results support the second hypothesis that plant responses vary across increasing but low foliar concentrations, highlighting the importance of application rate in determining PH effectiveness.

Taken together, the results demonstrate that foliar-applied PHs can enhance plant physiological performance, particularly in cherry tomato, and that their effects are strongly dependent on both formulation and application rate. These findings support that PH efficacy is crop-specific and dose-dependent, supporting their potential role in sustainable and resource-efficient agricultural systems. The limited or inconsistent effects on biomass accumulation may be attributed to the fact that PHs primarily induce rapid physiological and metabolic adjustments (e.g., photosynthesis, stomatal regulation, and nitrogen metabolism), which do not necessarily translate into immediate changes in structural biomass during the vegetative stage. As biomass is a cumulative trait, longer time periods may be required for these physiological improvements to result in measurable dry matter differences. Future research should therefore evaluate PH effects over the full crop cycle and under field or stress conditions to determine their impact on yield and long-term productivity.

5 Conclusion

Foliar application of plant- and microbial-derived PHs enhanced vegetative growth, photosynthetic performance, and water-use efficiency in cherry tomato, whereas responses in common bean were more limited and primarily associated with improvements in intrinsic water-use efficiency. Low application rates (0.05–0.1%) were sufficient to induce these responses, highlighting the high efficiency of PHs at reduced doses.

These findings demonstrate that PH effectiveness is both species-specific and dose-dependent, emphasizing the need to tailor biostimulant strategies to individual crops. Overall, PHs show strong potential as sustainable tools for improving crop performance under controlled conditions, although further studies are required to confirm their effects on yield and productivity under field conditions.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44279-026-00771-5>.

Supplementary Material 1.

Author contributions

F.B, T.H.N and T.G conceptualised the research. F.B. conducted the experiments, collected and analysed the data, and wrote the original manuscript. T.H.N. and T.G. supervised the research, contributed to the experimental design and interpretation of results, and provided critical revisions to improve the manuscript. All authors reviewed and approved the final version of the manuscript.

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Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable. This study involved only greenhouse cultivation of plants and did not involve human participants, animals, or clinical samples.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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